

PITTSBURNT 2.6

Developer Change Spec

2.5 implementation review plus approved 2.6 changes - slower truthful playback, spatially distributed tree siting, experienced UTCL, compact panels, and a Gemini-assisted custom-solution workflow.

Locked decisions: run the complete 2,409-segment / 167.16-km modelled network; keep Street Tree + Shaded Waiting Shelter as the two reliable built-ins; use a balanced-protection default; compare 2026 baseline, future before Adjust, and the same future after Adjust. Custom solutions are an optional Beta layer.

What changes in 2.6

- Empty start: no selected time, population, scenario, result, people, thermal tint, hotspot, or budget panel.
- People first: neutral moving agents appear as soon as Who + Time are selected and Moving People is enabled. Crash Test recolours them by computed UTCL and adds exposure halos.
- No false cliff: 38 C UTCL remains the published entry to Very Strong Heat Stress, while the visual ramp and optimizer use continuous heat load above 26 C.
- Balanced Adjust: first protect the highest waiting-exposure unsheltered stop on each Hero corridor, then spend the remainder by deterministic marginal benefit.
- Exact placement: one map icon per purchased unit, actual shade footprints, and post-intervention UTCL arrays returned by the engine.
- Friendly demo UI: larger type, floating rounded cards, explanations on hover/focus, a large centre-bottom Crash button, and a post-crash Adjust modal.
- Full reset: clear React state, Map feature-state, agent data, placements, timers, overlays, and camera state.
- Reproducible build: repair the lockfile and lint error before handoff.
- Legible motion: use one 15x demo time-compression factor; a 150 m route takes an Older Adult about 11.1 s and a Student about 7.7 s on screen.
- Spatial siting: replace segment-capacity point placement with 8-10 m candidate sites, overlap-aware marginal benefit, and deterministic minimum spacing.
- Experienced UTCL: show a person-minute-weighted before/after value; keep unchanged air temperature neutral, not green.
- Add New Solution (Beta): Gemini researches and structures evidence, asks for missing assumptions, and explains the result; the deterministic engine alone computes placements and impacts.

Truth rule

Animate only values on the path to a validated snapshot. The language model may retrieve and structure sourced ranges, ask questions, and explain a plan, but it never invents or directly commits costs, effects, placements, or impact numbers.

1. Verified 2.4/2.5 audit - what is actually wrong

Priority	2.4/2.5 implementation	2.6 correction
P0	Defaults are hard-coded to 2035 / 3 PM / Older adults / \$250K / All.	Make selection fields nullable. Disable Crash until Time + Who + Scenario are complete.
P0	Moving agents require a Crash result and the people reveal stage.	Render neutral agents before Crash after Who + Time. Bind heat colour only after a snapshot exists.
P0	Every agent completes a route in 14 s, regardless of persona speed.	Use real route minutes with one explicit demo playback multiplier. Zoom never changes ground speed.
P0	Reset leaves hotspot feature-state and the last agent source data.	Clear all feature-state keys and empty every dynamic GeoJSON source; cancel timers and animation frames.
P0	Future summary colours the first N time boxes, not the actual crossing hours.	Remove the summary from the main UI; if retained in evidence, store exact crossing_hours.
P0	After-map sun is inferred from an exposure ratio.	Return after_sun, after_utci_c, after_shelter_coverage, and shade geometry directly from the engine.
P0	All 300 cached Across-all plans buy zero shelters.	Replace pure ROI default with the approved balanced-protection policy, clearly labelled as a policy constraint.
P1	A placement is one green dashed segment even when it contains many units.	Return point geometry per unit and render exact tree/shelter counts with staggered icon drops.
P1	SOU / ASS / COM badges are hover-only title attributes.	Make them buttons that open, scroll, focus, and highlight the matching source section.
P1	Panels use 10-12 px text and a 310 px + 350 px dense layout.	Increase demo-critical labels to 16-24 px; move detail to accessible popovers.
P0	npm ci fails because package-lock is out of sync; lint has 1 error.	Regenerate and commit lockfile; fix BudgetField effect; run clean-install CI.

Checks completed: the UTCI engine property suite passes 42/42. A production build succeeds after dependency repair. Full pipeline verification was not executed in this review environment because GeoPandas is not installed; keep it as a release gate.

2. State machine, empty start, and full reset

Start with a context map, not a pre-answered question. Existing protection layers may be toggled independently, but no analytic state is implied until the user makes a selection.

Nullable selection contract

```
interface Selection {
  scenario: ScenarioId | null;
  hour: 8 | 12 | 15 | 18 | null;
  persona: PersonaId | null;
}

initial = {
  selection: { scenario: null, hour: null, persona: null },
  result: null, adapted: null,
  crashStage: 'idle', adaptStage: 'idle',
  layers: { shadow:false, canopy:false, trees:false,
            buildings:false, trips:false, agents:false }
};

canRunCrash = Boolean(scenario && hour && persona);
showBudget = crashStage === 'complete';
```

Interaction flow

State	Visible UI	Primary action
Empty	Context map + three large selectors + optional protection layers	Crash button disabled with “Choose 3 inputs”
Ready	Neutral moving agents if enabled; no red streets or result card	Large red RUN CRASH TEST
Crash complete	Climate / UTCI / Human Exposure result; top hotspot	Crash button morphs to ADJUST
Adjust modal	Budget input + Balanced Protection default + concise method tooltip	BUILD THE PLAN
Adjusted	Exact placements + cooling / protected-wait bloom + three-stage result	REPLAY or RESET

Reset acceptance contract

```
reset():
  cancel crash/adapt controllers and all requestAnimationFrame handles
  abort pending API calls
  set selection and results to null
  clear segment feature-state: scored, utci, severe, sun, hotspot
  empty agents, placements, shade-footprints, trails GeoJSON sources
  close/minimize overlays and remove map popups
  restore context-only camera and layer defaults

assert(dynamicSourceFeatureCount === 0)
assert(noSegmentFeatureStateIsActive)
assert(noTimerOrAnimationIsRunning)
```

3. Layers and moving people

The two tree layers answer different questions. Rename them so the difference is visible without explanation.

Control label	Meaning	Verified data	2.6 visual
Existing Tree Inventory	Individual public tree points from the City inventory	2,231 points	Cluster at overview zoom; individual crowns on zoom-in
Existing Canopy Coverage	Overhead foliage raster, including vegetation absent from the point inventory	2010 vintage; 32.3% model area / 36.3% context	Soft green texture below points; show vintage on hover
Moving People	A deterministic visible subset of synthetic routed trips	50 visible agents from the selected cohort	Neutral before Crash; UTCI colour + cumulative halo after Crash

Movement implementation

```
physicalDurationMs = route.minutes * 60_000;
const DEMO_TIME_COMPRESSION = 15;
visualDurationMs = physicalDurationMs / DEMO_TIME_COMPRESSION;
u = ((now - agent.startMs) % visualDurationMs) / visualDurationMs;

// Simulation truth remains in metres and seconds.
// Zoom changes pixels-per-metre only; never speed_mps or route.minutes.
// Use the same 15x time-compression for every persona.
// It changes display time only. Exposure calculations retain physical time.
```

Persona speeds - planning assumptions

Persona	Physical speed	Required visual test
Students	1.30 m/s	Same route appears 44% faster than Older adults
Workers	1.20 m/s	Same route appears 33% faster than Older adults
Older adults	0.90 m/s	150 m: 166.7 s physical -> 11.1 s visual at 15x
Mobility-constrained	0.80 m/s	Slowest cohort at every zoom level

Playback acceptance: remove the 2-second floor that makes short routes look equally fast. For the same 150 m route, the Student must take about 7.7 s and the Older Adult about 11.1 s. Cumulative severe dose remains monotonic after the walker enters shade.

4. Climate reveal, UTCI, and Human Exposure

Do not animate humidity, wind, or solar radiation upward. The current scenario method changes air temperature only; RH, wind, and radiation are held constant to isolate the warming delta. Animate temperature and the derived Tmrt / UTCI / heat burden, and label the controls as held constant.

Verified default story - 3 PM, Older adults

State	Air temp	RH / wind / solar	Experienced UTCI	Heat load	Severe exposure
2026 hot-day baseline	33.05 C / 91.5 F	39% / 2.97 m/s / 765 W/m2	35.30 C	12,238	0 person-min
2035 before Adjust	35.04 C / 95.1 F	held constant	37.79 C	15,525	949.83 person-min
2035 after current \$250K plan	same future weather	held constant	37.47 C (-0.32 C)	15,101.37	838.05 person-min
2050 before Adjust	36.62 C / 97.9 F	held constant	39.80 C	18,162	949.83 person-min

The 2035 and 2050 3 PM severe totals are equal because the binary threshold counts the same sun-exposed minutes once both scenarios cross 38 C. Therefore the main animated burden counter should be continuous Heat Load; show Severe Heat Exposure as the published-class count beside it.

The 2035 full-sun and full-shade anchors remain 38.85 C and 35.06 C. “Future After” must not show the text “local by segment.” Show the person-minute-weighted Experienced UTCI above; show a clicked segment's numeric local UTCI in the map popup.

Metric contract

```
Human Exposure = walking exposure + waiting exposure

experienced_utci =
  SUM(utci_i * person_minutes_i) / SUM(person_minutes_i)

// For waiting minutes, use the sheltered/unsheltered UTCI actually experienced.
// Recompute after every plan; never average unweighted street segments.

severe_person_minutes =
  SUM(person_minutes in sun where UTCI_sun >= 38 C) +
  SUM(person_minutes in shade where UTCI_shade >= 38 C)

cumulative_heat_load =
  SUM(max(UTCI - 26 C, 0) * person_minutes)

Map A: absolute UTCI band on the street
Map B: human hotspot halo / agent dose
Never use exposure rank to recolour the thermal band.
```

2035 / 3 PM band check

Scope	Active segments	32-38 C Strong	38-46 C Very Strong	Inactive / context
Full modelled network	579	260	319	1,830 of 2,409
Hero corridors	68	24	44	88 of 156

“1 of 4 hours” means one of four sampled times, not one hour of a four-hour period. Remove it from the main screen with Across the day. If retained in Sources, display exact sampled times and actual crossing hours.

5. Balanced Adjust - Tree + Shaded Waiting Shelter

The current optimizer is behaving consistently with its inputs, but not with the desired planning story. Waiting is only 3.4% of the 2035 / 3 PM severe exposure, a tree costs \$1,200, a shelter costs \$15,000, and pure severe-minutes-per-dollar therefore chooses trees in all 300 cached Across-all plans.

Approved objective and spatial candidate layer

Candidate generation - applies to every scenario and budget
sample each eligible footway every 8-10 m using stable candidate IDs
reject existing canopy/building shade and infeasible geometry
enforce minimum spacing; one footprint cannot be counted twice

Phase A - transparent service floor
For each Hero corridor:
select the eligible unsheltered stop with the highest
computed waiting exposure and place one shaded shelter.

Phase B - deterministic, overlap-aware marginal allocation
With the remaining budget, repeatedly choose the feasible site
with the greatest NEW severe-person-minutes avoided per dollar;
after every placement, recompute nearby candidates so shade overlap
and already-protected person-minutes have zero duplicate benefit;
break ties by heat-load avoided, cost, then stable candidate ID.

Label: Balanced Protection policy
Do not label it a pure global optimum.

Verified \$250K default arithmetic and simulation

Plan	Units / spend	Severe exposure	Trade-off
Current pure efficiency	208 trees, 0 shelters / \$249,600	949.83 -> 820.47 (-13.62%)	Best current total reduction; no transit protection
Approved Balanced Protection	170 trees, 3 shelters / \$249,000; \$1,000 left	949.83 -> 838.05 (-11.77%)	Protects one highest-wait site per Hero corridor

The balanced result above is a regression fixture from dataset version 6a0955d7 under the pre-2.6 segment-fraction model. The 2.6 candidate-site rebuild must recompute and replace every count and impact. Do not hard-code 170 + 3; the budget, eligibility, spacing, scenario, persona, costs and evidence profile determine the answer.

Service-floor sites for the verified default

Hero corridor	Selected segment ID	One-shelter waiting severe minutes avoided
Forbes Avenue	mckee-place-0431-13	0.4930
Fifth Avenue	fifth-avenue-0556-01	1.0859
South Craig Street	south-craig-street-1183-00	0.4778

Evidence limitation: stop service frequency is not stop ridership. The model assumes 20% of walking trips end in a transit wait (10-30% sensitivity range). Present the shelter service floor as a planning policy choice, not as an unconstrained AI discovery. Candidate locations are planning sites, not construction-ready engineering locations; utilities, pits and right-of-way clearance remain unavailable.

6. Adjust payload and exact map choreography

The UI must not reverse-engineer cooling from an exposure score. The engine owns all post-intervention environmental arrays and exact point placements.

Required response fields

```
interface UnitPlacement {
  unitId: string;
  kind: 'tree' | 'shaded_shelter';
  segmentId: string;
  stopId?: string;
  lon: number;
  lat: number;
  costUsd: number;
  order: number;
}

interface AdaptResult {
  placements: UnitPlacement[]; // one record per purchased unit
  afterSun: number[];
  afterUtciC: number[];
  afterShelterCoverage: number[];
  shadeFootprints: FeatureCollection;
  before: ExposureSnapshot;
  after: ExposureSnapshot;
  policy: 'balanced_protection';
  inputHash: string;
}

assert(placements.length === treeCount + shelterCount);
```

Ten-second Adjust animation

Time	Beat	Map / panel behaviour
0.0-1.0 s	Lock	Freeze budget, policy, scenario, agents, and input hash.
1.0-5.0 s	Place	Drop bundled emoji-style tree and shelter sprites in optimizer order; counts and spend tick to exact totals.
5.0-7.0 s	Protect	Tree shade blooms along footways; shelter roof bloom stays at WAIT_NODE. Do not cool the whole road for a shelter.
7.0-9.0 s	Re-test	Replay identical agents and routes; agent colour / halo follows the returned after arrays.
9.0-10.0 s	Land	Show 2026 baseline, future before, future after; place Replay and Reset.

Use bundled SVG or PNG emoji-style sprites instead of operating-system emoji glyphs. This preserves the requested friendly look while keeping symbol size and appearance deterministic on the demo machine.

Tree icons must use the optimizer-selected candidate coordinates, not $(n + 0.5) / \text{segmentCapacity}$. For presentation, units selected on one continuous street component stay at least 8 m apart and distribute across their purchased-site extent. Shelters use exact validated stop coordinates. Cluster only after the drop animation; the result panel always shows the exact count.

7. Demo-first UI redesign

Use the reference images as layout principles: map-first composition, soft floating surfaces, generous spacing, large controls, and data-bound cooling blooms. Do not copy their content or decorative heat spots.

Area	2.6 requirement
Left setup card	About 332 px (roughly 12% smaller than 2.5). Keep large selectors; move detail to hover/focus. Header has Minimize and Restore.
Primary CTA	Centre-bottom circular red button, at least 96 px, strong label and warning pulse. Disabled until three required inputs exist.
Post-crash Adjust	Do not show budget initially. Morph the Crash CTA to Adjust; open budget and policy in a focused overlay.
Right result card	About 356 px. Air Temperature is neutral “same weather”; Experienced UTCI, Heat Load and Severe Exposure show numeric before/after.
Overlays	Every overlay has Minimize and Restore; dismissible overlays also have Close. A minimized panel becomes a labelled pill. Panels remain fixed while the map moves.
Responsive gate	Nothing clips at 1366x768 or 1920x1080. Use max-height: calc(100dvh - 32px) and internal scrolling only when expanded.
Provenance badges	SOU / COM / ASS / MIS are buttons. Click opens Sources, scrolls to data-key, focuses it, and applies a 1.5 s highlight.

Remove from the default demo surface

- Across the day and the confusing “N/4 hours” summary.
- Top-five hotspot list - show the top one and a View all drawer.
- Planning weights, raw synthetic trip counts, segment counts, and model disclaimers - move to Sources.
- Long descriptions under every selector, live-engine status text, and intervention unit-cost detail before Adjust.
- Dense multi-row legend - keep one compact UTCI gradient with 26 / 32 / 38 / 46 markers.

Crash choreography - 7.5 seconds

0.0-1.5 Climate: 91.5 F -> selected future temperature
RH / wind / solar remain visibly LOCKED
1.5-3.0 Thermal: Tmrt and UTCI rise; gradient crosses bands
3.0-5.5 People: neutral agents recolour; exposure dose accumulates
5.5-7.5 Hotspot: top human-exposure location pulses and result lands

Counters finish on exact snapshot values even if frames drop or Skip is used.

Judge-facing demo contract - implement all six

#	Moment	Required visible behaviour
1	Empty start	Only Who, When, Future Scenario and the disabled Crash CTA. No results, budget, people or heat layer.
2	Crash story	Run the 7.5 s climate -> thermal stress -> people -> hotspot sequence; never land all results in one tick.
3	People first	Show 30-50 legible representative agents. Recolour and pulse people before strengthening the road overlay. Label the represented trip count.
4	Three headline metrics	Air Temperature, Experienced UTCI and Severe Exposure only. Move Heat Load and technical detail into Details/Sources.
5	CTA handoff	After Crash, morph the centre CTA from CRASH TEST to PROTECT THIS CORRIDOR and only then reveal budget/policy choices.
6	Ten-second Adjust	Scan candidates -> place exact Tree/Shelter icons -> bloom local protection -> replay identical agents -> land corridor and network impact.

Zoom Easter egg - human-scale cause and effect

- Entry: enable only after Crash at map zoom >= 16. Pointer and keyboard focus reveal that red agents, planted Trees and Shaded Shelters are interactive.
- Red agent click: pause that representative agent, add a 1.2 s sweat-droplet animation and show its route time, local UTCI band and accumulated exposure. This is a modelled trip, not an individual medical diagnosis.
- Tree/Shelter click: highlight the unit's actual protection footprint. The nearest on-screen red representative agent within 50 m follows a short preview path to the protected point; sweat fades and its colour transitions to the computed after-UTCI band.
- No fake green: use green only when after UTCI is below 32 C. If the intervention only reaches Strong or Moderate stress, stop at orange or yellow. Shelter interaction applies to waiting exposure at its stop; it does not cool the entire street.
- Non-destructive: the 3-4 s Easter-egg preview never mutates routes, counts, optimizer output or headline metrics. On close/replay it returns to the exact deterministic snapshot.

8. Add New Solution (Beta) - Gemini-assisted, engine-controlled

Keep Street Tree and Shaded Waiting Shelter as the two deterministic built-ins. Add New Solution is an optional Beta after the core demo is locked: Gemini may research, translate and question; it never computes or commits impact.

Responsibility boundary

Component	Allowed	Forbidden
Gemini API	Search grounded evidence; extract a typed draft; ask 1-3 missing-field questions; explain the optimiser trace	Invent an effect, choose coordinates, alter engine arrays, or claim medical benefit
Validation API	Check units, ranges, required fields, source URLs, confidence and supported mechanism	Silently convert an unsupported claim into a cooling factor
Deterministic engine	Generate eligible sites; calculate UTCI/exposure deltas; optimise under budget; return exact placements	Use free-form model text as a numeric parameter
User	Confirm planning cost, operating assumptions, behaviour and low-confidence overrides	Turn an assumption into sourced evidence by clicking confirm

Typed solution draft

```
interface SolutionDraft {
  name: string;
  geometry: 'point' | 'linear' | 'area';
  mechanism: 'solar_block' | 'tmrt_modifier' |
    'microclimate_node' | 'wait_or_rest' | 'access_only';
  eligibleSite: string;
  unitCost: { capexUsd:number; annualOpexUsd?:number;
    sourceStatus:'sourced'|'user_assumption'|'unsupported' };
  effect: { parameter:string; value:number; unit:string;
    radiusM?:number; operatingHours?:number[] };
  behaviour?: { dwellMinutes?:number; participationRate?:number };
  sources: { url:string; claim:string; retrievedAt:string }[];
  confidence: 'high' | 'medium' | 'low';
  confirmedByUser: boolean;
}

if required evidence is missing: status = 'cannot_simulate';
if supported and confirmed: register adapter, then run deterministic optimize();
```

Chat sequence

Step	Example behaviour
1. Describe	User: “Add misted rest shelters near busy stops.”
2. Research	Gemini uses Google Search grounding and returns linked cost/effect ranges, never one unexplained number.
3. Clarify	Ask whether cost includes plumbing/O&M; whether it protects waiting only or adds a voluntary rest; what radius/hours have evidence.
4. Confirm	Show SOURCED / ASSUMPTION / UNSUPPORTED badges. User approves the exact planning inputs.
5. Simulate	Backend validates the schema; deterministic siting and exposure engine compute the plan.
6. Explain	Gemini summarizes the trace and limitations; all displayed numbers come from the returned snapshot.

Deployment: call Gemini only from FastAPI `/solution-chat`; keep `GEMINI_API_KEY` server-side and model name configurable. If Gemini or search fails, retain the two built-ins plus cached example drafts. The Crash - Adjust - Re-test loop must remain fully local-first.

9. External heat-policy catalogue and model fit

Other cities validate a broader solution library, but not every policy can honestly change pedestrian UTCI in this street model.

City / programme	Observed policy direction	Pittsburnt treatment
Phoenix - Shade Phoenix	Plan calls for more than 27,000 trees and 550 built shade structures.	Tree remains built-in; built shade becomes a linear solar-block custom template.
Miami-Dade - Extreme Heat Action Plan	Trees, cool commutes, shade/overhangs, hydration, misters, splash pads and cool-pavement pilots.	Separate physical shade, microclimate, rest behaviour and access-only benefits.
Los Angeles	Federal funding announced for cool pavement and shade structures.	Shade fits now; pavement requires a measured Tmrt/surface profile.
New York City - CoolRoofs	Reflective roof coating reduces building heat and cooling demand.	Useful policy reference but out of the pedestrian-corridor MVP.
Paris - OASIS schoolyards	Vegetation, permeable surfaces and shade create area-scale cooling/refuge spaces.	Requires an area/refuge and optional dwell model; do not fake as a street shade multiplier.

Supported mechanism gate

Candidate	2.6 status	Required numeric evidence
Shade sail / awning / arcade	SIMULATE NOW	Solar-block fraction, shaded length/area, geometry and cost
Cool pavement	GUARDED TEMPLATE	Measured Tmrt or radiative parameters by hour; surface cooling alone is insufficient
Misted shaded rest node	NEW ADAPTER	Air-temperature/RH delta, radius, hours, cost, water/O&M; and wait/dwell rule
Hydration station only	ACCESS METRIC ONLY	Service radius and access count; no UTCI or heat-load reduction claim
Cool/green roof	OUT OF STREET MVP	Building-energy/longwave coupling not present in this engine

Cool-pavement caution: Phoenix measured lower pavement surface temperature but higher midday mean radiant temperature over the reflective treatment. A generic “cool pavement = lower pedestrian UTCI” factor is therefore rejected.

Sponsor/API fit

MLH lists a Best Use of Gemini API prize and an MLH-Google Gemini partnership. Gemini supports function calling, structured JSON output and Google Search grounding, so it fits the research-and-validation chat. MLH guidance says it cannot provide Gemini usage credits; use an AI Studio key and verify the active quota/pricing.

Primary references

City of Phoenix / ASU Cool Pavement study | Miami-Dade Extreme Heat Action Plan | MLH prizes | Gemini Search grounding | Gemini structured output | Gemini function calling

10. Release gates and developer handoff

P0 - must pass before the next demo

Gate	Pass condition
Empty start	No selected pills, no result, no people, no thermal/hotspot layer, no budget panel.
People toggle	After Who + Time, neutral agents appear without running Crash.
Speed truth	15x display compression only; 150 m is about 11.1 s for Older Adults and 7.7 s for Students; exposure keeps physical seconds.
2035 verification	At 3 PM / Older adults: 579 active = 260 Strong + 319 Very Strong; no percentile recolouring.
Reset	After Crash or Adjust, zero dynamic features and zero active feature-state remain.
Adjust truth	Exact icon count equals payload count; shelter changes waiting coverage, not street sun. Tree candidate points are at least 8 m apart.
Balanced default	Recompute the \$250K result after candidate-site rebuilding. Never hard-code 170 trees + 3 shelters or old impact values.
Experienced UTCI	Air temperature stays neutral; person-minute-weighted UTCI is numeric before/after and agrees with local popup values.
Custom solution Beta	Unsupported effects cannot run. With Gemini offline, built-ins and cached drafts still complete the core demo.
Easter-egg truth	At zoom ≥ 16 , red-agent sweat and protection previews work by pointer and keyboard. Colour uses computed after UTCI; preview changes no engine state or metric.
Three-state result	2026 baseline -> selected future before -> same future after; never compare 2026 directly to adjusted future as intervention effect.
Build	Fresh npm ci, npm run lint, npm run build, 42 engine tests, pipeline verify, and deterministic replay all pass.

P1 - polish after P0

- Viewport-safe minimize/restore behaviour and keyboard-accessible tooltips.
- Icon clustering after placement; exact unit detail on zoom or click.
- Smooth camera easing to the top hotspot and back to the full network.
- Reduced-motion path that keeps the same ordering and exact final values.

Known data limitations that must stay visible in Sources

Input	Status / limitation
Tree \$1,200; shelter \$15,000	Low-confidence planning assumptions, not Pittsburgh procurement quotes.
20% transit-trip share	Low-confidence assumption; expose 10-30% sensitivity.
Walking trips / persona mix	Synthetic fixed-seed samples, not calibrated pedestrian counts or Census shares.
Canopy	2010 raster; may omit recent planting and removals.
Building heights	59.5% measured or sourced; remainder modelled with record-level confidence.
Future weather	CMIP6-LOCA2 temperature delta applied to observed hot-day profile; RH, wind, and radiation held constant.
Tree feasibility / growth	Current capacity is segment-length based and mature shade is immediate; utilities, pits, survival, and Year-1 growth are not yet modelled.
Shelter demand	Service frequency proxy plus assumed trip share; no stop-level boarding calibration.

Final product sentence

Pittsburnt does not promise a medically safe street. It shows where a reproducible future heat scenario overlaps with human walking and waiting, then makes the policy trade-off behind a budgeted shade plan visible.